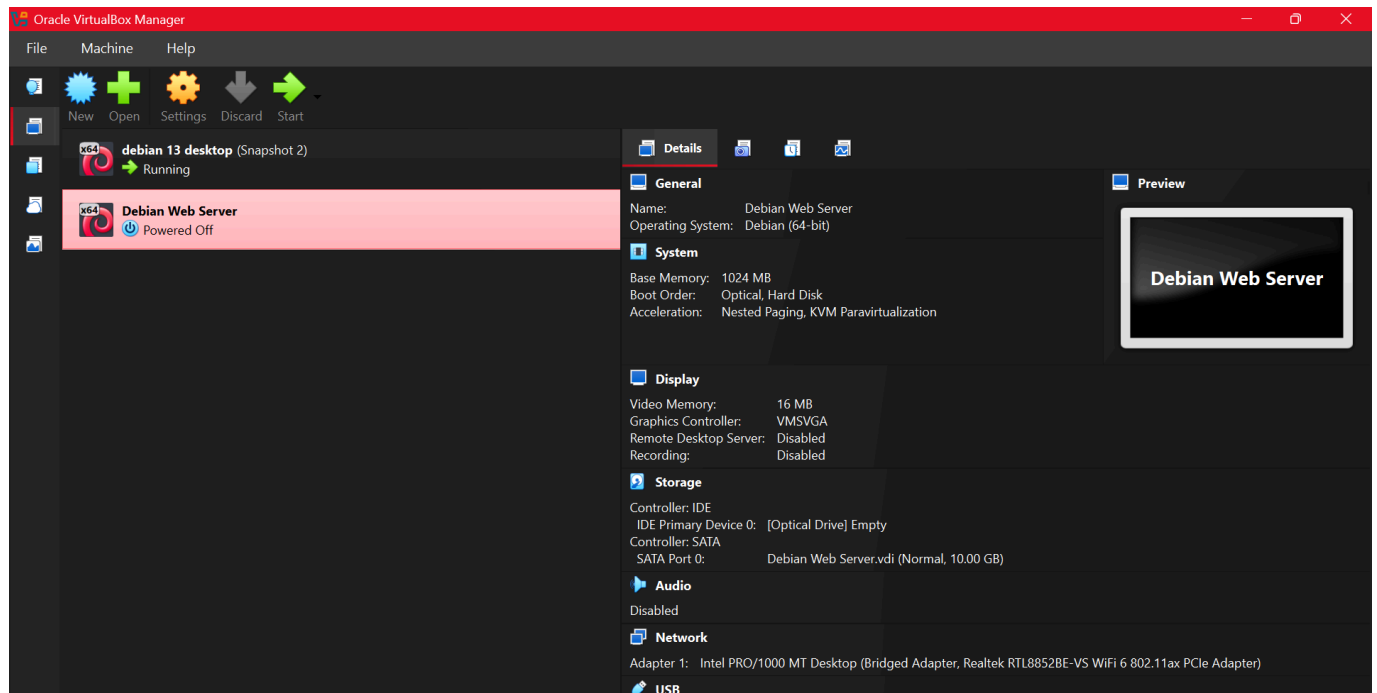
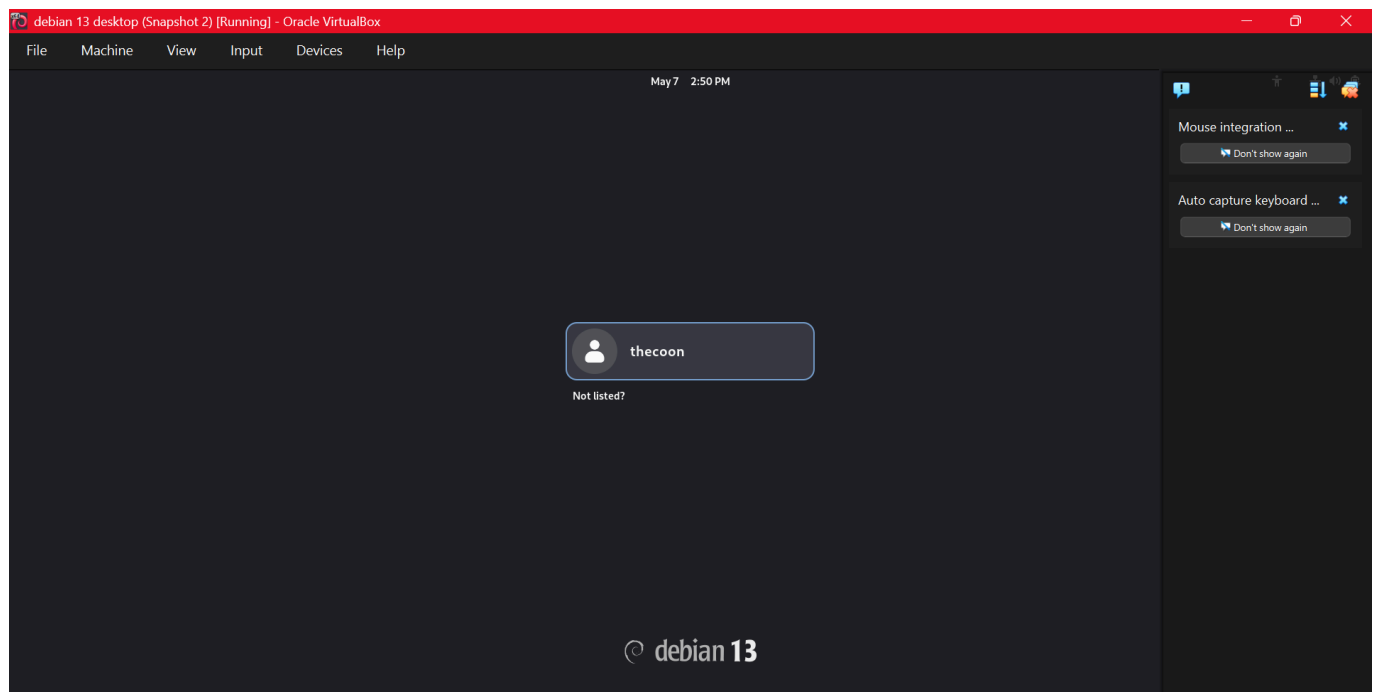


## Deliverable 2

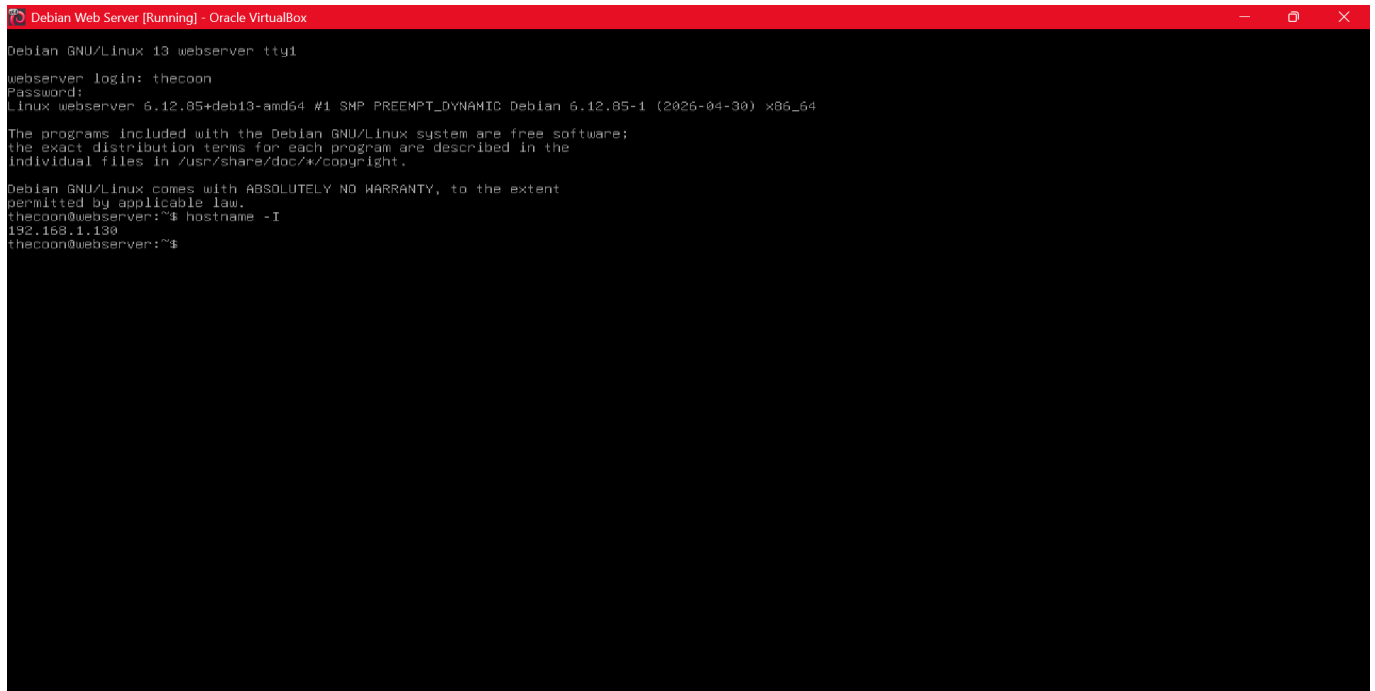
What are the server hardware specifications (virtual machine settings)? Take a screenshot - don't type it!



What is the Debian Login Screen? (Take screenshot - do not type it!)



What is the IP address of your Debian Server Virtual Machine? (type the command and show a screenshot of the commands output)



```
Debian GNU/Linux 13 webserver tty1
webserver login: thecoon
Password:
Linux webserver 6.12.85+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.85-1 (2026-04-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
thecoon@webserver:~$ hostname -I
192.168.1.139
thecoon@webserver:~$
```

### How do you work with the Firewall in Debian? (Type and explain what each command does)

`sudo ufw enable` - Used to enable a service. `sudo ufw disable` - Used to disable a service. `sudo ufw status` - Shows the status of a program. `sudo ufw allow ssh` - Prevents you from getting locked out of your VM

### How do you check if the Firewall is running?

- To check if the Firewall is running, run `systemctl status ufw`
- This tells you whether it is active or not

### How do you disable the Firewall?

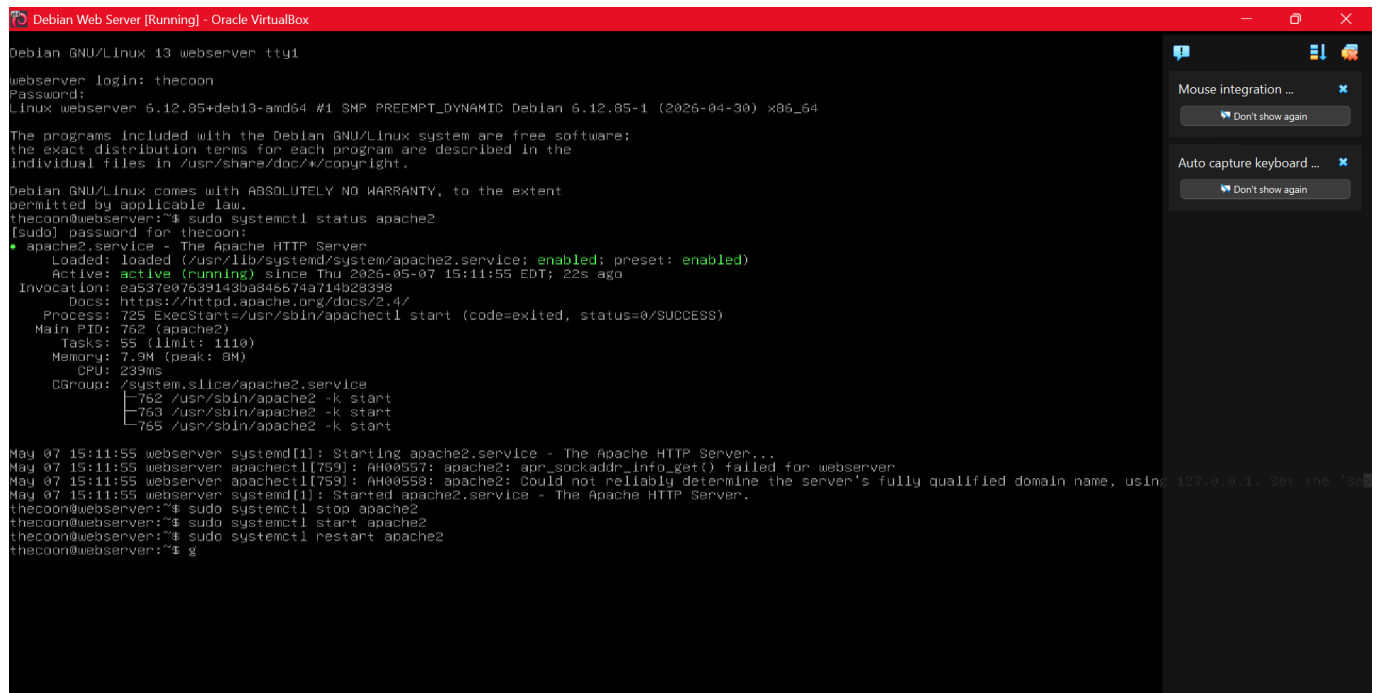
- To disable the Firewall, run `sudo ufw disable`
- After running, `systemctl status ufw` will show the status as inactive.

### How do you add Apache to the Firewall?

- To add Apache to the Firewall, run `sudo ufw allow 'Apache'`
- This allows Apache to serve web pages to anyone on the network.

### What different commands do we use to work with Apache? (Type and explain what each command does and include a screenshot!)

- `sudo systemctl status apache2` - Shows whether Apache is running or stopped.
- `sudo systemctl start apache2` - Starts the Apache web server.
- `sudo systemctl stop apache2` - Stops the Apache web server.
- `sudo systemctl restart apache2` - Restarts Apache.



```
Debian GNU/Linux 13 webserver tty1
webserver login: thecoorn
Password:
Linux webserver 6.12.85+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.85-1 (2026-04-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
thecoorn@webserver:~$ sudo systemctl status apache2
[sudo] password for thecoorn:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-07 15:11:55 EDT; 22s ago
     Invocation: ea537e07639143ba846674a714b28398
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 725 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 762 (apache2)
      Tasks: 55 (limit: 1119)
     Memory: 7.9M (peak: 8M)
        CPU: 239ms
     CGroup: /system.slice/apache2.service
            └─762 /usr/sbin/apache2 -k start
              └─763 /usr/sbin/apache2 -k start
                └─765 /usr/sbin/apache2 -k start

May 07 15:11:55 webserver systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 07 15:11:55 webserver apachectl[759]: AH00557: apache2: apr_sockaddr_info_get() failed for webserver
May 07 15:11:55 webserver apachectl[759]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1. Set the 'ServerName'
May 07 15:11:55 webserver systemd[1]: Started apache2.service - The Apache HTTP Server.
thecoorn@webserver:~$ sudo systemctl stop apache2
thecoorn@webserver:~$ sudo systemctl start apache2
thecoorn@webserver:~$ sudo systemctl restart apache2
thecoorn@webserver:~$ g
```

### What is the command you use to check if Apache is running?

- To check if Apache is running, run `sudo systemctl status apache2`
- This tells you whether it is active or not

### What is the command you use to stop Apache?

- To stop Apache, run `sudo systemctl stop apache2`

### What is the command you use to restart Apache?

- To restart Apache, run `sudo systemctl restart apache2`

### What is the command used to test Apache configuration?

- To test Apache Configuration, run `sudo apache2ctl configtest`
- This tests all configuration files for any errors

### What is the command used to check the installed version of Apache?

- To check the installed version of Apache, run `apache2 -v`
- This shows you the current version and the build date

### What are some common configuration files for Apache?

- Some common configuration files for Apache are:
  - `apache2.conf`
  - `ports.conf`
  - `httpd.conf`

### Where does Apache store logs?

- Apache stores all of its logs in */var/log/apache2/*

**What are some basic commands we can use to review logs?**

- **Some basic commands to review logs are:**
  - *cat* - Displays the contents of a file.
  - *tac* - Displays the contents of a file in reverse order.
  - *head* - Displays the first *N* lines of a file.
  - *tail* - Displays the last *N* lines file.